工程數學 Final Exam January 10, 2023

ID:_____ Name:_____

滿分 20 分, 每題 4 分

(1)

$$f(t) = \begin{cases} 1, & 0 \le t < 4 \\ 0, & 4 \le t < 5 \\ 1, & t \ge 5 \end{cases}$$

(2)

$$L^{-1}\{\frac{(1+e^{-2s})^2}{s+2}\}$$

(3)

$$F(s) = \frac{e^{-2s}}{(s-1)^3} \qquad Find\ f(t)$$

(4) find b_n

$f(x) = x + \pi\ ,\ -\pi < x < \pi \quad and \quad f(x+2\pi) = f(x)$

(5) find the Fourier series of the function f on the given interval

$$f(x) = \begin{cases} 2+x, & -2<x<0 \\ 2, & 0 \le x < 2 \end{cases}$$

公式表:

- Laplace Transform

	$f(t)$	$\mathcal{L}(f)$		$f(t)$	$\mathcal{L}(f)$
1	1	$1/s$	7	$\cos \omega t$	$\frac{s}{s^2+\omega^2}$
2	t	$1/s^2$	8	$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$
3	t^2	$2!/s^3$	9	$\cosh at$	$\frac{s}{s^2-a^2}$
4	t^n $(n = 0, 1, \cdots)$	$\frac{n!}{s^{n+1}}$	10	$\sinh at$	$\frac{a}{s^2-a^2}$
5	t^a (a positive)	$\frac{\Gamma(a+1)}{s^{a+1}}$	11	$e^{at}\cos \omega t$	$\frac{s-a}{(s-a)^2+\omega^2}$
6	e^{at}	$\frac{1}{s-a}$	12	$e^{at}\sin \omega t$	$\frac{\omega}{(s-a)^2+\omega^2}$

First Shifting Theorem, s-Shifting

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$$

$$e^{at}f(t) = \mathcal{L}^{-1}\{F(s-a)\}$$

Laplace Transform of Derivatives

$$\mathcal{L}(f') = s\mathcal{L}(f) - f(0)$$

$$\mathcal{L}(f'') = s^2\mathcal{L}(f) - sf(0) - f'(0)$$

$$\mathcal{L}(f^{(n)}) = s^n\mathcal{L}(f) - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - f^{(n-1)}(0)$$

Laplace Transform of Integral 積分轉換

$$\mathcal{L}\left\{\int_0^t f(\tau)\,d\tau\right\} = \frac{1}{s}F(s), \quad \text{thus} \quad \int_0^t f(\tau)\,d\tau = \mathcal{L}^{-1}\left\{\frac{1}{s}F(s)\right\}$$

Second Shifting Theorem; Time Shifting

$$\tilde{f}(t) = f(t-a)u(t-a) = \begin{cases} 0 & \text{if } t < a \\ f(t-a) & \text{if } t > a \end{cases}$$

$$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$$

$$f(t-a)u(t-a) = \mathcal{L}^{-1}\{e^{-as}F(s)\}$$

Examples for t-shifting by 1

$1 \mapsto \frac{1}{s}$ $\qquad u(t-1) \mapsto e^{-s}\frac{1}{s}$

$t \mapsto \frac{1}{s^2}$ $\qquad (t-1)u(t-1) \mapsto e^{-s}\frac{1}{s^2}$

$t^2 \mapsto \frac{2}{s^3}$ $\qquad (t-1)^2u(t-1) \mapsto e^{-s}\frac{2}{s^3}$

$\sin t \mapsto \frac{1}{s^2+1}$ $\qquad \sin(t-1)u(t-1) \mapsto e^{-s}\frac{1}{s^2+1}$

傅立葉級數 *(週期 2L)*

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi}{L}x + b_n \sin\frac{n\pi}{L}x\right)$$

(a) $a_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\,dx$

(b) $a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx \qquad n = 1, 2, \cdots$

(c) $b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}\,dx \qquad n = 1, 2, \cdots$

Fourier Cosine Series (*even function, 偶函數*) *(週期 2L)*

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi}{L}x \qquad (f \text{ even})$$

$$a_0 = \frac{1}{L}\int_0^L f(x)\,dx, \qquad a_n = \frac{2}{L}\int_0^L f(x)\cos\frac{n\pi x}{L}\,dx, \qquad n = 1, 2, \cdots$$

Fourier Sine Series (*odd function, 奇函數*) *(週期 2L)*

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi}{L}x \qquad (f \text{ odd})$$

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}\,dx$$

$\sin\pi = 0, \quad \sin 2\pi = 0, \quad \sin 3\pi = 0, \ \ldots$

$\cos\pi = -1, \quad \cos 2\pi = 1, \quad \cos 3\pi = -1, \ \ldots$